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Agent-based simulation applications in marketing research: an integrated review

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Agent-based simulation has been used to study a wide variety of complex and social systems. This paper presents the results of a comprehensive and multi-dimensional analysis of the application of agent-based modelling and simulation in marketing. Papers are evaluated with respect to types of interaction and communication channels, individual cognitive decision-making styles, social system structures, application area, goals of the simulation study, and other model properties. Furthermore, network analysis and text-mining methods are employed as two novel approaches to establish the current state of the literature and reveal literature trends, gaps, and strengths. Various emergent phenomena and important findings from the application of agent-based simulation in different areas within marketing are also discussed. Finally, the results of the literature analysis are used to make future research recommendations by providing an architecture for conceptual model design and general guidelines to increase the credibility of agent-based models in the marketing domain.

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1. Introduction

Computer simulation can be defined as the process of developing a model of a real or hypothetical system for the purpose of conducting experiments to understand its behaviour. The use of simulation is expanding as a result of its successful applications in many real-world problems. Traditionally, simulation techniques have been widely used to analyse complex stochastic systems ranging from manufacturing (Smith, 2003), healthcare (Mielczarek and Uzialko-Mydlikowska, 2012), supply chain (Terzi and Cavalieri, 2004), business (Jahangirian *et al*, 2010), military (Naseer *et al*, 2009), and engineering (Crowder *et al*, 2012) systems. During the past few decades, social sciences have become a fertile area for the application of simulation. In this context, agent-based modelling and simulation (ABMS) has been used for several purposes, mainly prediction, proof, and theory development (Axelrod, 1997). For example, ABMS is used to prove Conway's Game of Life by showing that simple rules at the individual level can result in very complex behaviours at the system level (Poundstone, 1985). As another example, ABMS studies of Schelling (1978) cultural model for spatial segregation based on preferences show that segregation can be self-emergent and that unlike the conventional perception that attributed the phenomenon to imposed restrictions, preexisting ethnic preference is a driving force for urban segregation. The applicability and contributions of ABMS as a powerful modelling tool for understanding and decision making in social sciences has been addressed in several other publications (Gilbert and Terna, 2000; Bonabeau, 2002; Gilbert, 2007; Hughes *et al*, 2012).

The general topic of agent-based modelling (ABM) has been discussed in several studies (Tesfatsion and Judd, 2006; North and Macal, 2007). A number of publications have also discussed technical aspects of ABMS including design of experiments, input and output analysis, verification and validation, and parameter calibration (Bankes, 2002; Sargent, 2005; Janssen and Ostrom, 2006; Macal and North, 2010). The growing interest in ABM has led to the development of several ABMS tools and software packages, namely, RePast (North *et al*, 2006), SWARM (Minar *et al*, 1996; Terna, 1998), Mason (Luke *et al*, 2005), and NetLogo (Lytinen and Railsback, 2012). For a complete list of ABMS packages, see Robinson and Ding (2010), Railsback *et al* (2006), and Perrone (2005). Moreover, there have been several survey papers on the application of ABM in a variety of fields. Table 1 provides a list of these review papers.

Recently, researchers and practitioners have shown considerable interest in the application of agent-based simulation in marketing. The consumer marketplace is a dynamic complex system where consumers, investors, manufacturers, sellers, distributors, and other types of entities interact with one another and their behaviour results in the emergence of different marketing phenomena and patterns (Holland and Miller, 1991). Interactions between these players and individual decision-making behaviour add a layer of complexity to marketing research that may not exist in other disciplines (Rosser, 1999). The key advantages of ABMS for marking research that are beyond the reach of conventional tools are as follows: (1) ABMS provides a bottom-up approach, in other words, the atomic model elements are agents. Therefore, population heterogeneity in terms of attributes

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Table 1 Review papers on ABMS

Reference	*Comments*
Samanidou *et al* (2007)	Provide a survey of agent-based models of financial markets
LeBaron (2006)	Survey research literature on ABM in finance
Heath *et al* (2009)	Establish the current practice of ABM in terms of year of publication, field of study, simulation software, purpose, and validation issues
Zenobia *et al* (2009)	Review the literature on the agent-based artificial markets as a venue for innovation diffusion research and provide guidelines to improve their viability and practicality
Kiesling *et al* (2011)	Outline existing modelling aspects concerning consumer adoption behaviour and social influence and address the strengths and limitations of ABM in the context of innovation diffusion
Jennings *et al* (1998)	Provide an overview of research in autonomous agents and multi-agent systems development and identify and relate key concepts and applications to one another
Tesfatsion (2002)	Identify eight research areas in agent-based computational economics (ACE) and provide an outline of the objectives and characteristics of the ACE methodology
Herbert (2006)	Review ABM research for the analysis of innovation and technology change dynamics

and decision-making processes, which has been a matter of discussion in marketing research for a long time (Chatterjee and Eliashberg, 1990), can be easily implemented; (2) It provides the means to explicitly model the structure of the consumer social network and thus allows for investigating the impact of social influences on the dynamics of markets; and (3) ABMS has the potential to explain complex non-linear marketing patterns observed in the real world, as it is capable of capturing emergent phenomena that arise from the micro-level behaviour of consumers and their interactions.

ABMS applications to marketing research have resulted in the discovery of several marketing phenomena. As an example, agent-based simulation was used to discover that, given certain purchasing behaviour for people and under certain range of conditions, a well-known phenomenon called 'lock-in' occurs, which suggests that after a short period of time, one of the competing products dominates the marketplace and this domination persists (Janssen and Jager, 1999). Although the phenomenon itself had been previously discovered, it was only with the aid of ABMS analysis that deeper insights, as discussed later in the paper, were gained on its dynamics.

As a result of its advantages over traditional aggregate techniques and its contributions to the field, ABMS has proven itself as a powerful tool for studying and analysing the dynamics of consumer marketplace. In this paper, we present a detailed analysis of the published literature on the application of agent-based simulation to marketing research. More specifically, the objectives of this paper are to: (1) outline the key components of agent-based models of artificial markets and characterize three main application areas by extending the existing classification schemes; (2) provide a multi-dimensional analysis of the literature to identify trends, gaps, and strengths by using innovative approaches such as network analysis and text mining; and (3) provide recommendations for filling the identified gaps by proposing a general architecture for agent-based models of markets and providing guidelines to increase their validity and credibility.

The remainder of the paper is organized as follows. The search strategy and data collection are described in Section 2. The results of our analysis on different aspects of the literature are presented in Section 3. Section 4 presents guidelines on general architecture and validity of ABM models in marketing. Finally, summary and conclusions are provided in Section 5.

2. Literature review methodology

2.1. Search strategy and selection of papers

The literature search was conducted using relevant keywords and combinations thereof. The entire search process generated approximately 11 200 papers. In order to collect the initial set of papers, we selected papers based on their title, abstract, and keywords, and further filtering was done by full text reviewing. Finally, reference chasing was used to include more papers in the list.

2.2. Data extraction

Three key elements of agent-based models are identified to characterize and evaluate the reviewed models: (1) type of interaction; (2) structure of the social system; and (3) consumer decision-making behaviour. The following three sections provide the typologies for these model components.

Type of interaction. The critical role of social interactions in the dynamics of markets has been recognized since the very early studies in the field. Social interactions, mainly external influences, such as promotions and mass media advertising, and internal influences, such as word of mouth (WOM), have been the core of fundamental models and principles in the domain of marketing (Arndt, 1967; Bass, 1969).

By reviewing the literature, three main types of communication have been distinguished: WOM, advertising activities, and human players/interactions. WOM, also known as the neighbourhood effect, involves activities in which consumers talk about a product to other agents and may influence their belief in a positive or negative way. Several researchers have studied the general operation and mechanisms of WOM empirically (Richins, 1983; Dellarocas, 2003; Leskovec *et al*, 2007). Advertising, on the other hand, is a form of communication through media, direct mail, the Internet, magazines, posters, or other types of mediums, which is used to influence the consumer behaviour (for a number of studies on different aspects of advertising, see

Horsky and Simon, 1983; Abernethy and Franke, 1996; Leong *et al*, 1998). One could also involve humans in the loop of simulation by developing an environment in which real human players can act as agents and engage in different decision-making processes during the course of simulation. By integrating human and artificial agents in the same environment, it is possible to conduct interesting experiments to mimic real-world markets.

Consumer decision-making behaviour. Consumer decision making is another fundamental factor that has been the centre of a huge body of the marketing literature. Consumers are constantly making decisions regarding the purchase of products. Therefore, understanding their decision-making process is the key to the success and survival of companies in today's highly competitive marketplace. As a result, numerous approaches have been developed to model the decision-making behaviour of consumers and identify important factors that affect their decisions (Olshavsky and Granbois, 1979; Bettman *et al*, 1991). In the context of agent-based simulation of markets, consumers are generally modelled as autonomous entities that make purchasing decisions based on predefined behavioural rules. We have identified six main types of decision-making processes used in the literature as follows:

- *Personal preference*: This type of decision making generally involves a preference-matching function using which choices are evaluated by calculating the degree of similarity between consumer's preferences and product attributes.
- *State-based/stage-based*: These models are similar to the susceptible–infected–recovered models that are widely used in disease propagation studies. Using these decision processes, consumer agents undergo several stages before making the final decision of adoption or rejection of the product. As an example of these state transitions, consider a scenario where a consumer agent that is initially unaware of the product, learns about the product from advertisement. It may then become a potential adopter, adopter, or rejecter based on the magnitude of positive and negative WOM.
- *Utility-based*: A utility function is used to evaluate product choices and choose the product with the maximum utility value. Utility functions are typically based not only on personal preferences but also on WOM and advertisement influence, susceptibility, budget constraint and so on that distinguish them from preference-matching functions.
- *Exposure threshold*: This type of decision making is similar to that used in threshold models, where the individual adopts the product only if the fraction of their neighbours that have adopted is greater than a critical threshold.
- *Past experience*: In these models, unlike personal preference/belief decision-making models, preferences are not fixed and can dynamically evolve over time, depending on the consumer's experience of a product after using it that requires the presence of some sort of learning process in the consumer agent's behavioural rules.
- *Random*: We call it random or probabilistic decision making if the consumer chooses a product randomly from the set of available choices.

Structure of the social system. The structure of the social system in which consumer interactions take place is another important component of ABMS models in marketing. Katz (1961) compares ignoring the structure of the social system when studying social interactions with studying blood circulation without considering the veins and arteries. This paper distinguishes three main types of structures used to model the social world: *network representation*, *spatial/physical representation*, and *ports/channels.*

Networks can be classified as either static or dynamic. While the connections in a dynamic network change in time, the structure of static networks is fixed. Within each of these two general categories, different strategies can be employed to determine the linkages among agents as follows:

- Random (Watts and Dodds, 2007): In a random network, there is a probability of p that an edge exists between two nodes.
- Regular network or lattice (Goldenberg *et al*, 2010): In this network, each node is connected to a particular number of neighbours.
- Small-world (Watts and Strogatz, 1998): This network is generated by starting with a regular network. In the next step, some edges are randomly rewired.
- Scale-free (Delre *et al*, 2010): The structure of a scale-free network is such that a few agents have many neighbours while most of the agents only have a few, that is, the probability for each node of having n neighbours decays according to a power law.
- Preferential network (Barabási and Albert, 1999): This is a special type of scale-free network that is based on preferential attachments. In these networks, when a new node i is added to the network, it is attached to node j with a probability that is proportional to the number of links already connected to node j.
- Empirical networks (Leskovec *et al*, 2007): Empirical or real-world networks can be developed based on data extracted from social networking sites such as *Twitter*.

Spatial representation deals with modelling the physical layout of the environment. Modelling the layout of a shopping centre or city in order to find the best location of posters is an example of this type of system representation. Another type of spatial structure used in the literature is based on the region the agents reside. In region-based models, an agent interacts with other agents that are in the same region with her. As an example, consider a grid in which in a given time step, a consumer only interacts with agents distributed less than a certain radius away from where it stands.

Finally, when using ports/channels, there is no network structure or spatial representation to support interactions among agents. In these models, a hub or port is a place where agents

submit their outgoing messages and other agents in the environment can get access to these messages by doing a query. Consider a firm that desires to target a specific category of consumers. This is done by posting a message to a hub (it can be a forum or Website) and the targeted agents in turn receive the message when they query the messages available in the corresponding hub.

3. Results of the literature analysis

In this section, a detailed analysis of the published research on ABM applications in marketing is provided. Using the findings from the review process, the trends in the literature in terms of application area, goals of the studies, and model properties are presented. Moreover, two innovative approaches, namely, a network-based analysis and a text-mining method, are also employed to identify literature gaps and strengths and to provide a comprehensive perspective on the area. Finally, emergent phenomena and other important findings derived from the application of ABMS in marketing are presented.

3.1. Application area

The paper identifies three main application areas for ABMS in marketing: (1) conventional markets; (2) digital markets; and (3) new product diffusion. The three primary application areas are identified based on two sources of information: marketing literature and the data extracted from the reviewed papers. Papers classified under the 'Conventional Markets' category address the application of ABM in classic markets in which consumers learn about products through traditional marketing channels such as catalogue sales, conventional advertising mediums, or simply walking into stores. However, by reviewing the marketing literature it becomes clear that there is a general agreement on the fact that as the Internet has developed into a robust and ubiquitous channel for e-commerce, digital markets have emerged that have substantial differences from comparable conventional markets in terms of certain characteristics (Brynjolfsson and Smith, 2000; Malaga, 2001).

In digital markets, consumers' are influenced by online product reviews, social networking Websites, blogs, forums, and other online venues. Therefore, the global reach of digital contents available on the Internet leads to a much more rapid movement of digital WOM and product-related information among a larger group of consumers. Furthermore, the scale limitations encountered in traditional markets do not apply to digital markets making them more efficient with respect to price levels, price elasticity, menu costs, and price dispersion (Smith *et al*, 2001). As a result of these differences, companies are still struggling with understanding knowledge-acquisition behaviours and buying patterns in digital markets. Therefore, analysing the application of ABMS in this area could be of interest to marketing researchers and practitioners.

The third application area deals with the initial phase of market penetration of new products that is crucial to their future success. The process of innovation diffusion often follows a gradual pattern where a small portion of consumers, known as innovators or early adopters, initially adopt an innovation after they learn about it through advertising, and over time the innovation spreads among more conservative consumers, also known as followers, through WOM and social contagion. There is an extensive literature on the dynamics of new product diffusion. The general topic of innovation diffusion has been addressed in several books (Brown, 1981; Rogers, 2003) and numerous analytical models for new product diffusion have been proposed since 1960s. Perhaps, the most commonly used model in the industry is the Bass model (Bass, 1969, 2004) that relates the timing of initial purchase of new products to the number of previous buyers. For a short list of analytical innovation and technology diffusion models, see Mahajan *et al* (1990).

While existing review papers address the applicability of ABMS in new product diffusion context (Herbert, 2006; Zenobia *et al*, 2009; Kiesling *et al*, 2011), despite the growing interest in digital markets, applications of ABM to digital and conventional markets are not explicitly distinguished. This paper extends the existing classification schemes to cover these two application areas that have been previously ignored. Table 2 compares the number of references under each application area. It is worth noting that because of the generality feature of simulation models and their ability to fulfil more than one purpose simultaneously, a number of papers could be included in more than one category. For those papers that do not fall neatly into one of these application areas, the most applicable category is selected.

A summary of the results from the literature review is provided in Figure 1 to present the distribution of different types of interaction, social system representation, and decision-making styles among the three application areas. Perhaps the most important observation is the number of papers that suffer from some level of incomplete description of the model, which has been also identified as an important deficiency in ABMS practices in other fields, namely, biology, ecology, social sciences, economics, traffic, and military (Heath *et al*, 2009). In the marketing domain, our results suggest that roughly 20% of the papers do not provide complete information regarding the decision-making behaviour or type of social network used. This could have an adverse effect on the quality of the publications as well as the credibility of ABMS among marketing experts, as it inhibits repeatability and replicability. Therefore, far more attention needs to be given to providing detailed model descriptions in the future.

Table 2 Agent-based simulation application areas in marketing

Class	*Number of references*	*Percentage*
Conventional markets	29	36
Digital markets	14	18
New product diffusion	37	46
Total	80	

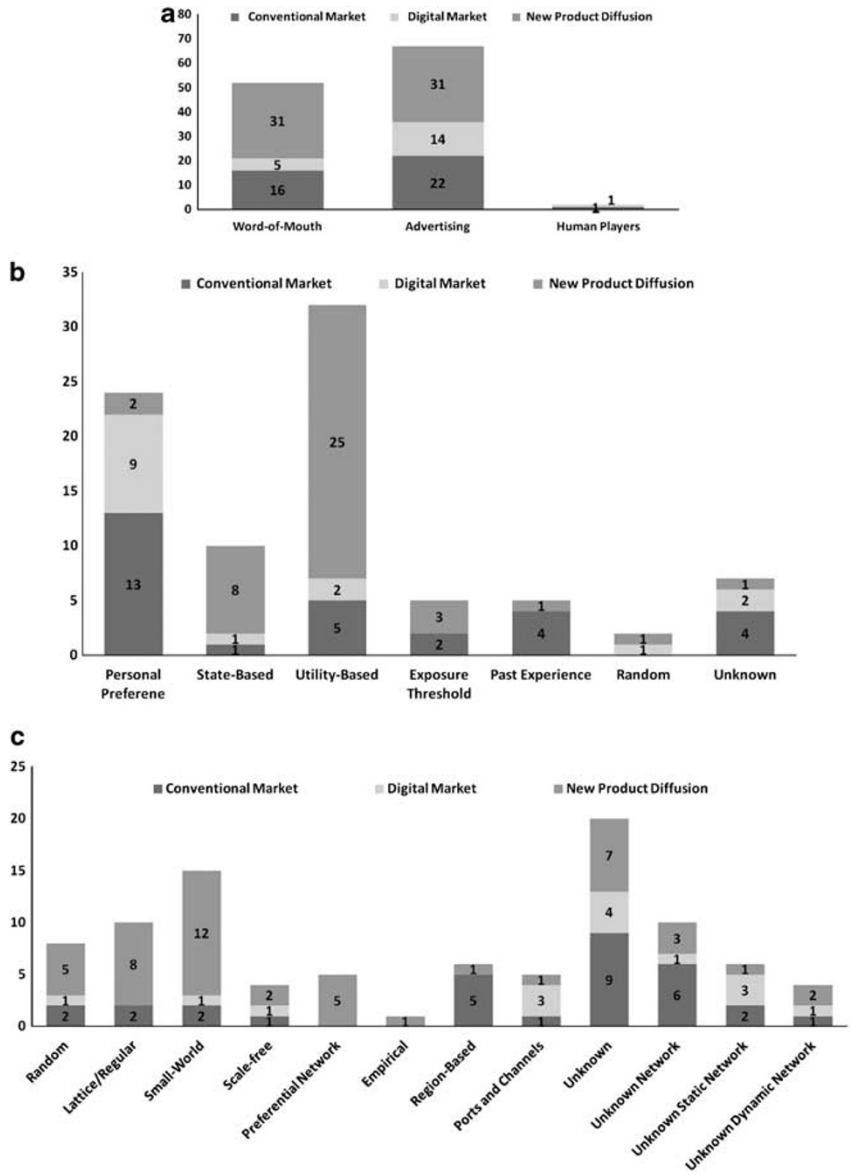

Figure 1 Distribution of model elements among the three application areas. (a) Distribution of interaction types among application areas; (b) Distribution of decision-making processes among application areas; (c) Distribution of different types of social system representation among application areas.

While WOM is considered in most of the papers under conventional markets and new product diffusion, only a few papers on digital markets consider WOM interaction among agents in the model. The reason may lie in our limited knowledge of the mechanisms and dynamics of the digital WOM, which provides an interesting avenue for future research. In the context of decision-making processes, utility-based and personal preferences are the most commonly used approaches among the three areas combined, whereas exposure threshold and past experience are among the least used methods. On the other hand, utility-based and state-based styles are found to be very common in diffusion studies, whereas decision making based on personal preferences and past experience are relatively more popular in studies related to conventional markets. In general, networks are found to be the most commonly used method to represent the social world. Small-world, random, and regular/lattice networks have been found to be the most commonly used network types in the three areas combined, whereas empirical networks, that are formed based on real data, have only been addressed in one paper. Small-world, regular, and preferential networks are relatively more popular in the realm of new product diffusion than conventional or digital markets. Finally, integration of human players into the simulation process has been studied in only two papers.

Agent-based models of markets can be categorized into two major streams as proposed by Kiesling *et al* (2011): (1) models

that present generic representations of the market under investigation. These studies are generic in the sense that their results do not correspond to a product and can be generalized and implemented in any study regarding the corresponding application area; (2) Models that are concerned with providing forecasts and decision support for a specific application. The structure of these models and/or their parameter configuration is typically exclusive to the particular product under study and thus most of the results cannot be generalized to other products. Table 3 provides the distribution of such papers under the three application areas.

3.2. Analysis of key model elements: a network perspective

We use the data collected on the three key model elements discussed in Section 2.2 and employ the Gephi network visualization package to generate a network representation of the use of these components in the literature (Figure 2). The size of a node is proportional to the frequency that the corresponding category is used and the width of edges is proportional to the number of times that the connecting nodes have been considered together.

We provide an analysis of different subnetworks to reveal research gaps and strengths in the literature (Figures 3–5). For instance, Figure 3(a) illustrates that small-world networks have been studied with most of decision-making styles except for random or past experience. Small-world networks have been also compared with most of the other social system representations except for region-based and dynamic networks. Using similar reasoning, a larger subnetwork could be an indication of a well-addressed topic, whereas smaller networks could represent the gaps in the literature. Similarly, the thickness of an edge could indicate how well the two nodes have been studied/compared together.

Table 3 Number (percentage) of papers in terms of scope among the three application areas

Goal	*Conventional market*	*Digital market*	*New product diffusion*
Generic studies	19 (66%)	9 (64%)	21 (57%)
Market/product specific studies	10 (34%)	5 (36%)	16 (43%)

Figure 2 A network representation of the literature.

Some important observations are summarized here. From the positive perspective, the impact of different decision-making processes has been addressed under various theoretical network structures such as small-world, lattice, and random. Moreover, Figures 3(a)–(f) illustrate that the existing papers provide an acceptable level of comparison between different network structures. With regards to decision-making modelling, Figures 4(a) and (b) suggest that the utility-based and personal preferences decision-making styles have been studied under different social system structures. On the other hand, as relatively more realistic network types, preferential and scale-free networks have not been studied along with personal preference, past experience, and state-based decision making. Most importantly, market dynamics under empirical networks have not been addressed under any of the decision-making style except for utility-based. On the other hand, dynamic networks that change with time are also more realistic than some of the theoretical network structures; however, they have not received much attention in the past. The difficulty in coding and implementation of these network structures could be the cause of their limited application. Finally, the lack of attention to incorporation of real human players is shown in Figure 5.

3.3. Goal of ABM studies in marketing

The publications fall into four general categories based on their goal: modelling consumer behaviour, advertising/promotion policy development/evaluation, verification and validation, and comparison of different simulation tools/techniques. The first category generally involves modelling consumer behavioural rules at the individual level and analysing the resulting macro-behaviour of the overall market consisting of heterogeneous consumers or groups of consumers. Therefore, these studies involve various research areas including psychology, sociology, and economics. Advertising/promotion policy evaluation, on the other hand, deals with implementing and testing different advertising and promotion strategies in order to provide managerial insights and policy recommendations. For a discussion on the advantages of ABM as a tool to evaluate economic policy design, see Dawid and Neugart (2011). This area also involves concepts and theories from other sciences such as psychology and economics. The third category generally involves the development or application of different techniques, methods, and approaches to make agent-based models more reliable, valid, and credible. The final category deals with the choice of modelling technique and comparison of different simulation tools and approaches.

Table 4 depicts the general trend of the identified study goals over the past two decades. The results suggest that consumer behaviour is the most active area. The significant growth in the number of papers since 2005 can be attributed to two main reasons. As a computationally expensive tool, this growth may be

Figure 3 Subnetworks for different types of social system structure. (a) Small-world network; (b) Random network; (c) Regular/lattice network; (d) Preferential network; (e) Scale-free network; (f) Empirical network; (g) Ports/channels social system; (h) Region-based social system, (i) Dynamic network.

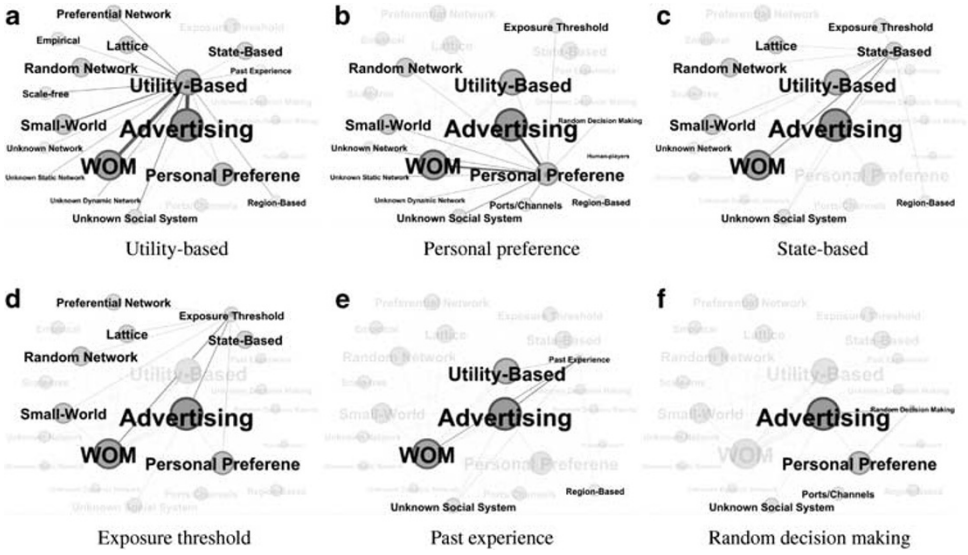

Figure 4 Subnetworks for different types of decision-making process. (a) Utility-based; (b) Personal preference; (c) State-based; (d) Exposure threshold; (e) Past experience; (f) Random decision making.

driven by the increase in the computational power of computers. The other reason could be the increased attention given to verification and validation in this period, which has led to increased credibility of this tool among both researchers and practitioners. Table 5 also presents the distribution of study goal categories by application area, which indicates the lack of attention given to verification and validation of agent-based models of digital markets.

Figure 5 Subnetwork for the human-player interaction type.

Table 4 Goals of the published research studies

Goal	*Number*	*Percentage*	*2000 and before*	*2001–2004*	*2005–2008*	*2009–2012*
Consumer behaviour	43	50	2	10	17	14
Advertising/ promotion policy	29	33.72	1	7	9	12
Verification and validation	5	6	0	0	4	1
Tools/techniques comparison	9	10.28	1	2	4	2
Total	86	100	4	19	34	29

Table 5 Distribution of study goals among three distinguished application areas

Goal	*Conventional market (%)*	*Digital market (%)*	*New product diffusion (%)*
Consumer behaviour	50	50	58.98
Advertising/ promotion policy	32.14	50	35.9
Verification and validation	3.57	0.0	2.56
Tools/techniques comparison	14.29	0.0	2.56

3.4. Model properties: number of products and outputs

In this section, we analyse the number of products in the model and simulation outputs/measures (Table 6). Number of products in the model and most of the subclasses of simulation outputs/measures are self-explanatory. The 'Other' subclass of simulation outputs includes consumer satisfaction, reputation, and advertisement annoyance. The results indicate that the majority of models in conventional and digital markets are multi-product, whereas most of the ABMS models under new product diffusion are single-product, which could be a limiting assumption. Finally, market share is by far the most commonly used output measure followed by profit.

Table 6 Agent-based model properties by application area

Property	*Conventional market*	*Digital market*	*New product diffusion*
Number of products			
Single	13	3	28
Multiple	16	11	9
Simulation outputs/measures			
Market share	25	12	31
Profit	4	2	3
Net present value	0	0	5
Rate of adoption	0	0	4
Other	3	0	2

3.5. A text-mining analysis of the literature

This section presents the results of a text-mining analysis as an innovative approach to identify important trends and characterize how the literature has evolved with time. We employ the RapidMiner open-source data mining package to extract the frequency of all phrases with up to three words from the full text of the 80 reviewed papers. Related words are then identified and clusters are formed to help us extract meaningful information. Finally, in order to be able to compare the usage of different word clusters through time, the frequencies are divided by the number of papers published in the corresponding time interval.

We focus on four aspects of the literature: (1) verification, validation, and calibration of agent-based models of markets; (2) use of network-based representations of the social world; (3) use of empirical input data and consideration of real-world problems; and (4) consideration of supply constraints. Table 7 provides the key words used to form these four clusters. The results of the text-mining analysis are provided in Figure 6.

The text-mining results on verification, validation, and calibration are consistent with findings from the literature review provided in Section 3.3. As shown in Figure 6(a), since 2005, there has been a surge in the use of words/phrases in this cluster. This increased attention to these issues can be attributed to the publication of several papers that specifically discuss verification, validation, and calibration of agent-based simulation models and raise awareness towards their importance (eg, see the papers by Rand and Rust, 2011 and Janssen and Ostrom, 2006). Similarly, the results from the text-mining analysis support our previous findings on the popularity of different network structures. Figure 6(b) illustrates the growing trend in the use of network-related terms/phrases and shows how networks have become the predominant method for modelling the structure of the social world in ABMS studies in marketing.

We were also able to gain two important insights from the text-mining results. The first finding is related to the use of empirical data, such as those used to develop empirical networks, and

Table 7 Word clusters used in the text-mining analysis

Verification, validation, and calibration	*Network representation of the social world*	*Empirical data and real-world implementation*	*Supply constraints*
Validation, valid	Network	Empirical	Supply
Validate(d)	Small-world, scale-free	Real-world problem(s)	Supplier
Verification, verify, verified	Preferential attachment	Case study	Capacity
Calibrate(d), calibration	Lattice, random network		Inventory

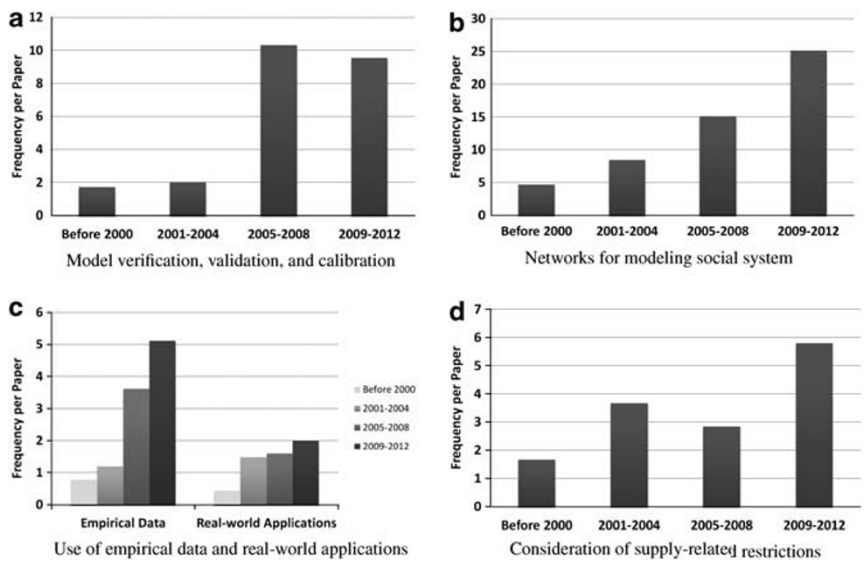

Figure 6 Text-mining results on ABMS practices in marketing. (a) Model verification, validation, and calibration; (b) Networks for modelling social system; (c) Use of empirical data and real-world applications; (d) Consideration of supply-related restrictions.

applications to real-world problems. Although our text-mining analysis indicates a growing pattern in this area (Figure 6(c)), the results from our literature review indicate that the majority of the publications in the field use a limited handful of theoretical network types. While these network representations are widely supported in the literature to provide useful results, marketing practitioners may find the assumptions of these networks unrealistic. Therefore, development of models that are based on empirical information and implementing their results in real-world settings is crucial to the adoption of ABMS by marketing practitioners in the future.

The other important finding is the growth in the consideration of supply-related restrictions in ABMS studies of markets (Figure 6(d)), which indicates a positive direction towards the development of more realistic models. However, based on our literature review, only a few studies have (implicitly) addressed supply-related restrictions, whereas the majority of ABMS studies essentially ignore these restrictions by assuming infinite supply or production capacity and zero manufacturing lead times. Therefore, far more attention needs to be paid to supply-related aspects in the future.

3.6. Emergence, important insights, and new findings

Emergence can be formally defined as a phenomenon that cannot be described solely based on the behaviour of its underlying components. In other words, emergent phenomena cannot be reduced to the constituent system parts and thus are difficult to predict or in some cases even difficult to understand when observed or discovered. The literature on social sciences provides numerous examples of counter intuitive emergent phenomena (Bonabeau, 2002). As a branch of social sciences, consumer marketplace is a complex system where the behaviour of many interacting agents and their ability to make decisions, learn, and react could lead to emergent phenomena that are difficult to predict or understand. This section highlights different emergent phenomena observed and outlines some other important insights derived from the application of ABMS in marketing research.

- *Lock-in dynamics*: Lock-ins typically happen when a small disturbance in favour of a product occurs in an emerging market and results in its dominance over the competitors. Lock-in effect examples include the QWERTY keyboard

layout and the competition between VHS and Betamax video tape formats. Through ABMS analysis of the emergence of lock-ins, it has been discovered that products that are relevant to customers' identity needs (the sense of belonging to a group) and quickly satisfy their needs are more likely to lock in (Janssen and Jager, 2001). Moreover, it is found that larger network size can cause fewer products to lock in. For the fashion market, it is found that it is optimal to introduce a new product when the incumbent is very popular, which seems counterintuitive. The importance of psychological factors is also discovered through simulation experiments. Furthermore, it is shown that under certain conditions regarding the formation of spatial regions within the population, it is possible to break a lock-in using strategies such as free give-away (Garlick and Chli, 2010).

- *Decoy effect*: Consider a market with two competing brands *A* and *B*, where *A* is superior to *B* in terms of quality but more expensive. Decoy effect occurs when a third brand *C* (also called a decoy), which is inferior to *B* in both quality and price, enters the market. In general, as *C* is inferior to *B*, customers do not adopt it. However, the mere introduction of *C* results in a substantial shift of preference from *A* to *B*, which is known as the decoy effect. ABMS results on the emergence of decoy effect, which is one of the most interesting phenomena in the marketing context, suggesting that customers with small difference between their motivation for brands *A* and *B* are more likely to switch, whereas this difference is large for the so-called loyal customers who never switch. Another discovery shows that whenever the price of decoy falls between the price of *B* and the average of price of *A* and *B*, the shift in the preference is much more explicit (Zhang and Zhang, 2007). Moreover, it is discovered that intensive WOM could decrease the loyalty of consumers and thus result in potentially more erratic market response to the introduction of decoy. Decoy effect indicates the importance of consumer psychology and their perception of product quality and price that makes ABMS an appropriate tool for studying this emergent phenomenon.
- *Breaking dominance*: Agent-based simulation has provided useful insights on the issues related to dominance in markets. ABMS is used to investigate the outcomes of forming alliances between smaller companies to compete with a competitor that holds a larger share of the market. It is found that despite the size (market share) disadvantage, under certain levels of network influence and distribution of market share, the alliance has the chance to succeed. Moreover, ABM results suggest that there is a non-linear relation between the network influence and the likelihood of alliance success (Chang *et al*, 2010). These findings are in contrast to the conventional view that believes size alone defines the nature of competition in the market and assumes a linear relation between network influence and alliance success. Such new insights, that in some cases contradict the traditional understandings, further indicate the significance of ABMS results and the important contributions of this tool to the marketing field.
- *Duopoly markets*: Several aspects of duopoly markets, where the consumer choice is only between two products, have been studied and interesting insights have been derived. Lock-in dynamics are studied with regard to duopoly markets as this type of emergent phenomena is commonly seen in these markets (Garlick and Chli, 2010). The impact of advertising on the dynamics of duopoly markets has been also investigated. As suggested in traditional studies, the stronger the advertising, the higher is the probability that the corresponding product will dominate the market. This is verified with simulation experiments. However, ABMS further reveals that under certain conditions, there is a critical level of advertising above which the dominance probability is 1 (Sznajd-Weron and Weron, 2003). This indicates that there is no additional benefit in increasing the advertising intensity above this critical level, which can help companies obtain the intended results with minimum advertising cost.
- *Role of influentials*: In the conventional marketing and diffusion research, it is widely accepted that influentials or opinion leaders play a significant role in the formation of public opinion. ABMS results support this understanding by showing that under certain conditions, influentials could have a considerable effect on diffusion processes by increasing the speed of adoption process and the adoption percentage (van Eck *et al*, 2011). However, the results from some other ABMS studies challenge this view by showing that the role of influentials could be less important than what was traditionally thought. These studies suggest that tracing the success of a marketing campaign to the individuals who started it or to the actions of a minority of special individuals is not an accurate interpretation (Watts and Dodds, 2007). Although ABMS studies reveal the shortcomings of the conventional view on the role of influentials, there is a need for more research and empirical studies to shed light on these contradictory results.
- *Role of cultural and socio-demographic factors*: Recently, agent-based models have been employed to investigate the impact of cultural and socio-demographic factors on the dynamics of markets. Although the importance of these factors has been recognized for a long time, it is only with the use of agent-based simulation in recent years that deeper insights on the function of these factors have been obtained. ABMS results indicate the difference in social influences from country to country (eg, see Broekhuizen *et al*, 2011). Such results can provide a significant aid to marketers to adjust their businesses according to the needs and characteristics of consumers in different countries and develop specific and effective marketing strategies based on cultural and socio-demographic differences to be successful in the international marketplace.
- *Impact of economic crisis*: More recently, and with the developments in the United States and Europe, agent-based simulation has been adopted to study the impact of financial crisis on consumer behaviour and dynamics of different markets. This has been found as a new application area for ABMS in the marketing context that can provide aid for marketers to survive in the volatile marketplace as a result of

a recession or any other disturbance. Although a few studies address such situations by investigating the performance of different marketing strategies as potential remedies to cope with tough market conditions (eg, see Johnson and Sieber, 2011), more work is yet to be done to fully utilize ABMS capabilities in providing insights for marketers to survive economic challenges during and after crises.

4. General guidelines for future practice of ABMS in marketing

On the basis of the results of our literature analysis and to address the identified gaps in the literature, this section aims at providing general guidelines for future practice of ABMS in marketing. We hope to achieve this by proposing an architecture for conceptual design of agent-based models of markets and providing guidelines that could potentially increase the validity of these models in the future.

4.1. A general architecture

An agent-based model of virtual markets consists of several types of entities including consumers, manufacturers or firms, retailers or distribution centres, sellers, products, services and so on. Interactions between these entities emerge in the context of the environment they reside. For example, social networks provide the setting for consumer or manufacturer agents to interact and exchange information. Each of these entities have attributes associated with them that define and describe their behaviour. Kiesling *et al* (2011) provide a structure for a typical agent-based diffusion model. However, in their proposed architecture, as in the majority the reviewed papers, only the consumer side of the market is considered and the supply network and its restrictions are completely ignored. The proposed architecture in this paper seeks to lay the foundations for more realistic models by explicitly illustrating the key elements of the supply chain network and the important interactions between the consumer and supply partitions. Therefore, such general structure provides the groundwork for future models that could potentially result in deeper insights on marketing dynamics, discovery of new marketing phenomena, and development of new theories. Figure 7 illustrates a proposed architecture along with examples of attributes or behavioural rules.

4.2. Validity of agent-based models

Verification and validation are two fundamental processes that are crucial to the development of simulation models of any kind. Verification is the process by which it is guaranteed that the model is right and corresponds to the conceptual model, whereas validation gives the modeller confidence on how accurately the model represents the real system, that is, that the right model is developed (Balci, 1994). This section provides a brief review of literature on model verification and validation and presents four areas for improvement specific to agent-based models in the marketing context.

Verification and validation of simulation models have been studied for years and various tools and techniques are proposed. Sargent (2005) provides 15 different validation techniques as well as techniques for data validity and conceptual model validation. They also discuss static testing and dynamic testing as the two basic approaches for verification of the computerized model. In the ABM context, Rand and Rust (2011) explain three important steps of verification through documentation, programmatic testing, and test cases. Moreover, they present four major validation techniques for ABM, namely, micro-face, macro-face, empirical input, and output validation. In another paper, Janssen and Ostrom (2006) discuss the use of empirical information to evaluate the validity of agent-based simulation models. For this

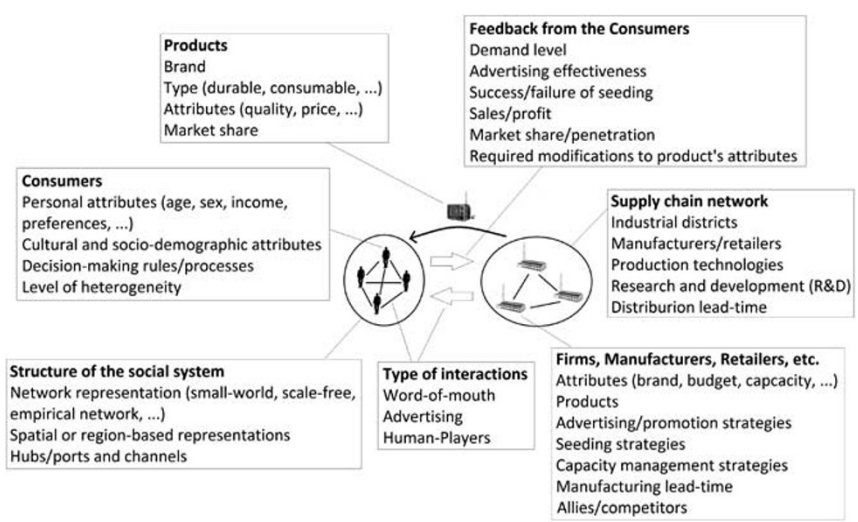

Figure 7 A general architecture for agent-based models of virtual markets.

purpose, they distinguish four approaches, namely, experimentation with uncomplicated models, conducting laboratory experiments under highly abstract and controlled environments, role play and companion modelling, and case-study analysis. The reader is referred to the original papers for more details on these verification and validation techniques.

The recommendations provided here are generally based on the results of our literature analysis. We believe that addressing the validity of these components can help resolve some of the existing scepticism towards ABMS applications in marketing (Leombruni and Richiardi, 2005) and increase the credibility and thus adoption of this tool by marketers and researchers. It should be noted that these recommendations should not be regarded as complete validation approaches, that is, it is still necessary to verify and validate models using the existing techniques in the literature.

- *Social network structure*: Today, there is a huge amount of information available about the structure of individuals' social network that was not available to researchers until a few years ago. ABMS could certainly benefit from the newly available database on social ties for transition from traditional and theoretical to empirical and more realistic social network structures. In order to improve social system representation in ABMS studies, we suggest the use of more realistic networks based on empirical data from various social networking Websites such as Facebook, Twitter, LinkedIn and so on.
- *Individual attributes*: Different attributes for consumer agents including age, gender, income, marital and housing status, and other personal or socio-demographic attributes have been recognized as important factors in agent-based models of markets. Traditionally, the value for each of these characteristics is assumed to fall within a predefined range and probabilistic distributions are then employed to specify the values when developing the virtual population of consumers. Although results from public surveys can be used to determine these ranges or percentages more wisely, marketing practitioners may dispute that the resulting population would not accurately represent the actual targeted group of potential customers due to the use of aggregate population data. Nowadays, with the advancement of data collection technology along with the significant growth in online marketing, companies have a wide spectrum of data regarding their current or potential customers that can be extracted using data mining tools and techniques. This will not only increase the validity of the model but will also help companies to fully utilize the potential of the micro-data available to them.
- *Individual decision making*: As a pivotal element of agent-based models in marketing, modelling consumer decision-making behaviour can have a significant impact on the validity of these models and thus plays an important role in the adoption of ABMS by marketing researchers and practitioners. However, current commonly used approaches for modelling individual decision-making process, as explained and classified earlier in the section 'Consumer decision-making behaviour', are mostly based on simple decision rules that has raised some scepticism from marketers regarding the validity of these models as well as the credibility of the results. Although we do not dispute the applicability of these methods and the significance of the insights derived from their application, we would argue that using more sophisticated and less parsimonious behavioural rules that are based on evolving psychological theories on behaviour will certainly help grow the acceptance of ABMS by marketing practitioners and researchers.
- *Incorporation of supply chain*: Besides supply constraints, there are other aspects of the supply network that are typically ignored in the existing studies. These include manufacturing or distribution lead times, demand control strategies of the firms, new product launch strategies such as build-up period policies, remanufacturing or refurbishing, production technologies, and product modification. The authors believe that considering both supply and consumption will certainly yield more realistic ABMS models and help grow the application of ABMS in marketing practice and research.

5. Summary and conclusions

We provide a comprehensive and multi-dimensional analysis of the published research on ABMS in marketing in order to identify literature trends, gaps, and strengths. On the basis of the literature analysis, the paper also provides guidelines and recommendations for future practice of ABMS in this area. The important findings and recommendations can be summarized as follows:

- *Literature trends*: The results show an increasing trend in the total number of publications in the field. The results also indicate a growth in the use of different network structures as well as the amount of attention given to model verification and validation. Growing patterns are also observed in the use of empirical data, applications to real-world problems, and consideration of supply chain restrictions.
- *Literature strengths*: Applications of ABMS in conventional markets and new product diffusion have been addressed in numerous studies. The role of advertising activities, WOM, theoretical network structures, and certain consumer decision-making styles have received a tremendous amount of attention in the literature. Moreover, the contributions of the results of ABMS studies to the marketing domain have proven the capability of this tool in capturing different emergent phenomena. While some of these results support the conventional findings, some other challenge the traditional/intuitive understandings.
- *Literature gaps*: The most important deficiency in the literature deals with incomplete description of the models in roughly 20% of the published research. Providing detailed information on the structure of the proposed agent-based models is crucial for improving the quality of the literature. As an emerging topic, the application of ABMS in analysing digital markets seems to be in its early stages. The operation of digital WOM, in particular, needs further investigation. Despite the growing

trend, only a handful of the reviewed papers use empirical network structures. On the other hand, the majority of the studies completely ignore supply-related constraints in their models. Finally, our analysis indicates that certain social system representation and decision-making styles have not been studied together.

- *Recommendations*: With the hope of reducing the identified gaps, the paper proposes an architecture for conceptual design of agent-based models of markets and presents general guidelines for increasing the validity of these models through the use of empirical social networks, real-world consumer profiles/attributes, more realistic decision-making styles, and consideration of different aspects of the supply chain network.

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